

VARUN HARIHARAN

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Aspiring AI Product Manager building AI-powered tools end-to-end — from problem scoping to deployed product — backed by a Human-Centered Design and Economics foundation.

EDUCATION

Northeastern University, Seattle Sep 2026 – May 2028

MS in Artificial Intelligence

University of Washington, Seattle Sep 2022 – Jun 2026

BS Human Centered Design & Engineering; BS Economics

PROJECTS

Monte Carlo Portfolio Risk Simulator Apr 2025 – Present

- Scoped and built an AI-powered risk analysis tool end-to-end, choosing to prioritize a natural-language input layer over a manual ticker/weight form to reduce the barrier for non-technical users
- Built a Cholesky-decomposed multi-asset Monte Carlo simulation with an AI-generated narrative layer (Claude API) that names the primary risk driver instead of returning raw output
- Backtested against 1,000 out-of-sample trading days, achieving a 4.60% breach rate against a 5% VaR target (Kupiec $p=0.56$) — confirming model calibration under normal conditions
- Stress-tested against the 2020 COVID crash window and identified a real limitation: on the worst single day, actual losses ran ~5.3x higher than the model's VaR forecast, motivating a fat-tailed distribution upgrade
- Deployed live on Streamlit Cloud

AI Data Analyst 2025

- Made a deliberate architecture decision to constrain the LLM to a fixed set of vetted analysis functions rather than free-form code generation, prioritizing reliability and safety over flexibility
- Built with FastAPI backend + Streamlit frontend, communicating over a REST API
- Stress-tested against a real 26,000+ row dataset; identified and resolved 4 distinct architecture issues (JSON serialization edge cases, a data-fabrication risk in the summary layer, column-type assumptions) through iterative testing

Research Clipper 2025 – Present

- Identified and scoped a citation-grounding gap in typical RAG tools, then designed a system that cites specific sources per claim and explicitly declines to answer when evidence is insufficient
- Validated grounding quality with a 16-question evaluation across direct-recall, negative-case, and cross-source-synthesis question types — 16/16 correct, including 3/3 negative cases correctly declined rather than hallucinated

EXPERIENCE

CoMotion Labs — Patent6 Quest (Undergraduate Capstone) Jan 2026 – Jun 2026

UI/UX Designer

- Made the core product decision to translate abstract IP/patent strategy concepts into gameplay mechanics, prioritizing intuitive understanding over strict accuracy to reach early-stage innovators
- Used playtesting data to identify which decision-flow friction points to fix first, informing a prioritized iteration roadmap

Valorant Registered Student Organization Jan 2025 – Apr 2025

UI/UX Designer

- Identified accessibility and navigation as the highest-leverage problems for a 1.5K+ member organization, based on synthesized findings from usability testing with 15+ users
- Prioritized 4+ rounds of iteration based on direct evidence of where users struggled, not aesthetic preference

PRODUCT & RESEARCH

Seesaw (Sep 2024 – Mar 2025) — Goal-tracking app for UW students; research with 25 participants, 8-user usability testing

City (Sep 2024 – Dec 2024) — Digital festival experience; branding, competitor analysis, mobile app concept

SKILLS

AI/ML: Claude API, Tool Use / Function Calling, RAG, Prompt Engineering, Sentence Embeddings, pgvector

Programming: Python, R, Java, JavaScript, HTML/CSS, React

Tools: FastAPI, Streamlit, Git, Figma, VS Code